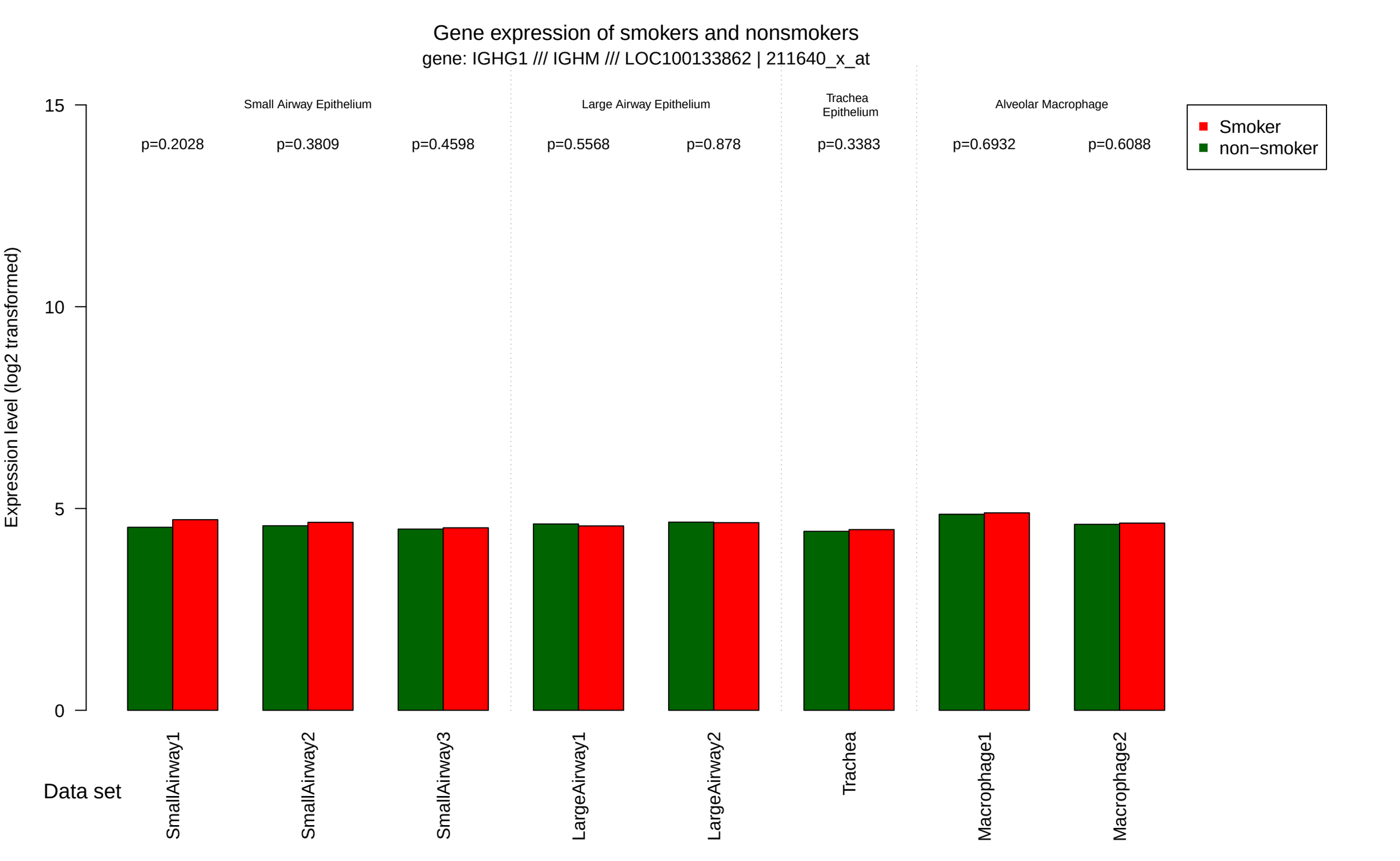
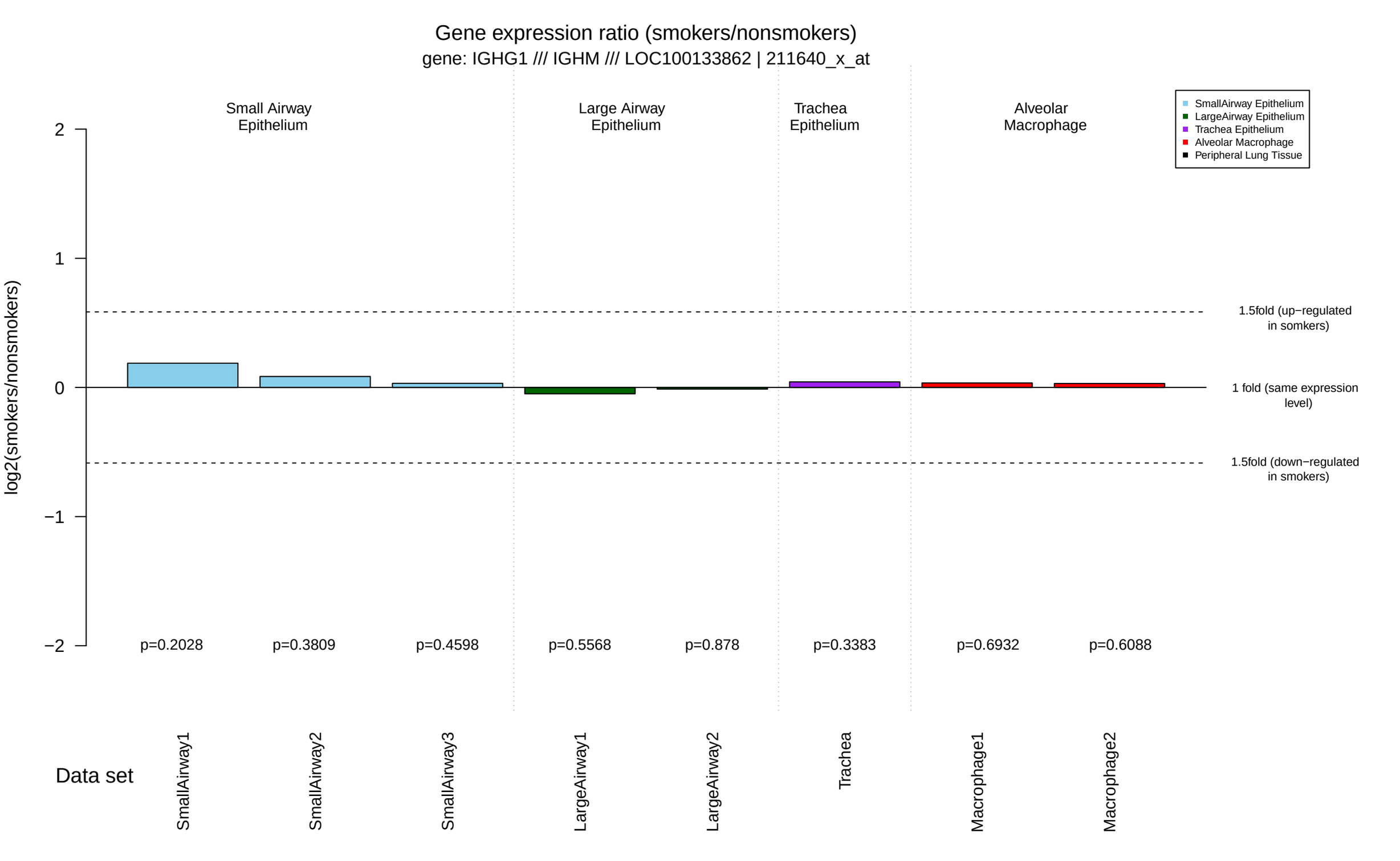
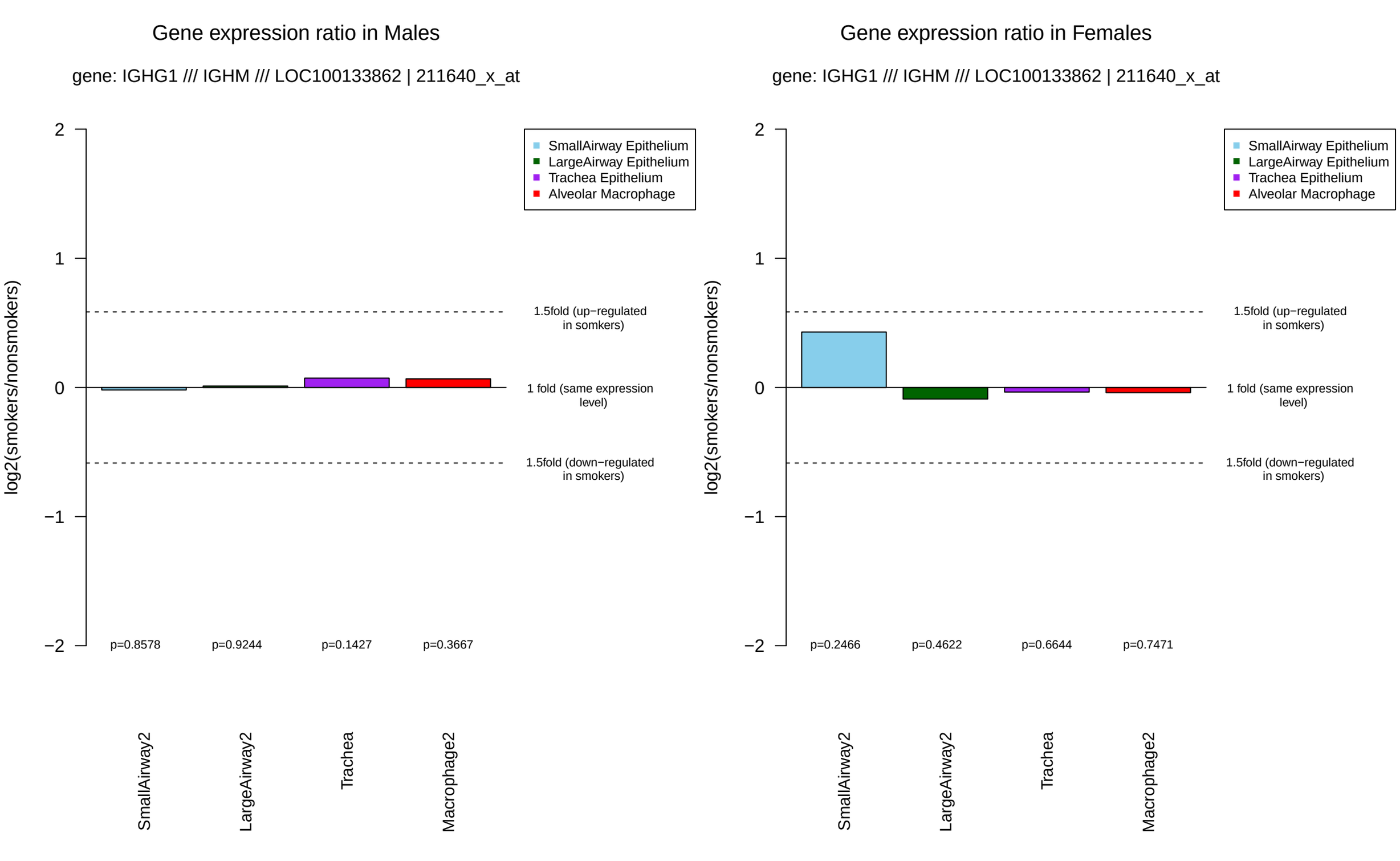
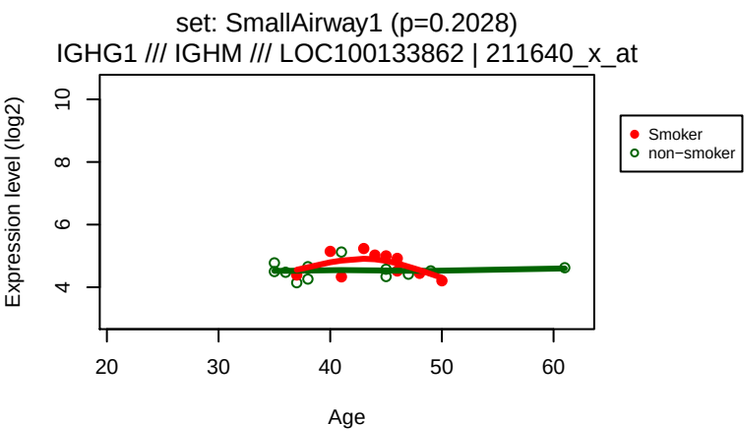
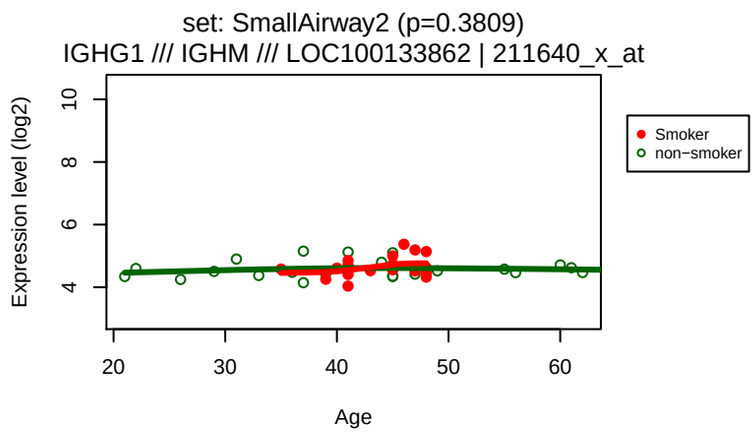
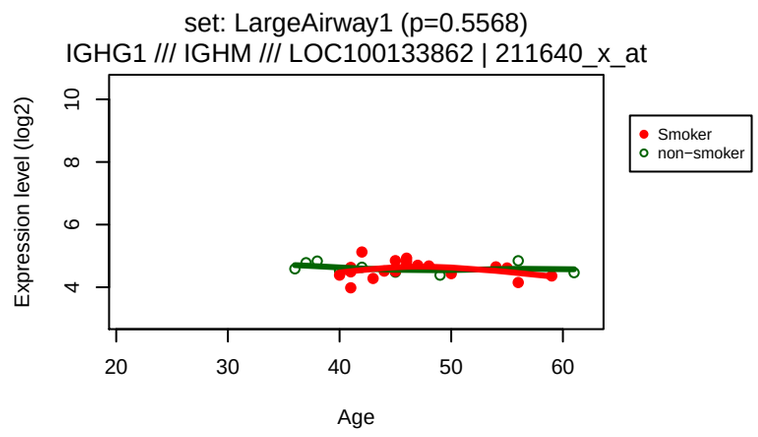

Gene expression of smokers and nonsmokers
gene: IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2028
p=0.3809
p=0.4598
p=0.5568
p=0.878
p=0.3383
p=0.6932
p=0.6088
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

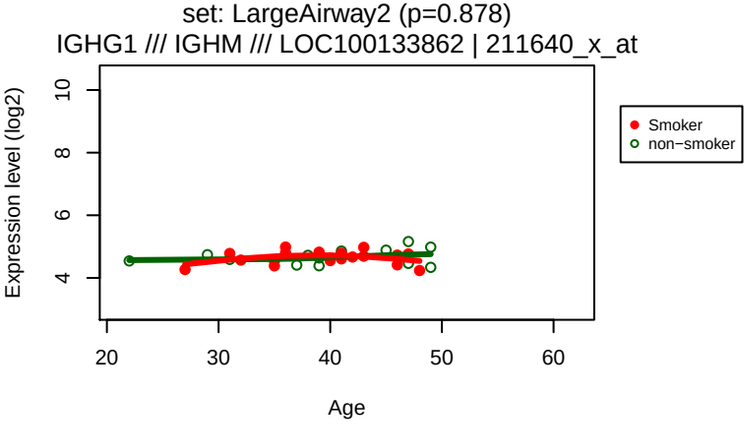

Gene expression ratio (smokers/nonsmokers)
gene: IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2028
p=0.3809
p=0.4598
p=0.5568
p=0.878
p=0.3383
p=0.6932
p=0.6088
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

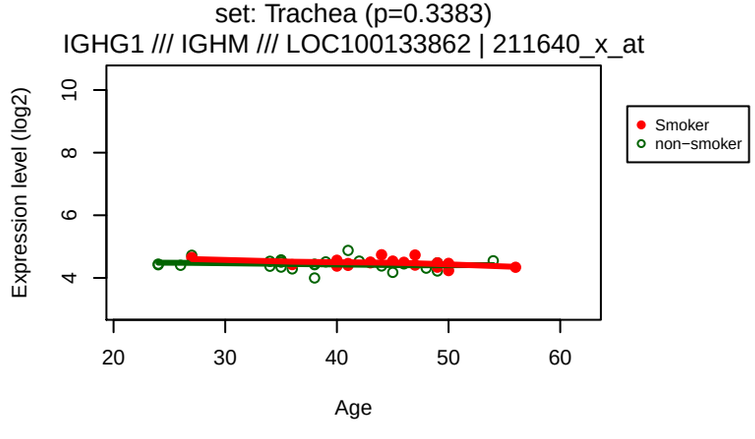

Gene expression ratio in Males
gene: IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8578
p=0.9244
p=0.1427
p=0.3667
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2466
p=0.4622
p=0.6644
p=0.7471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

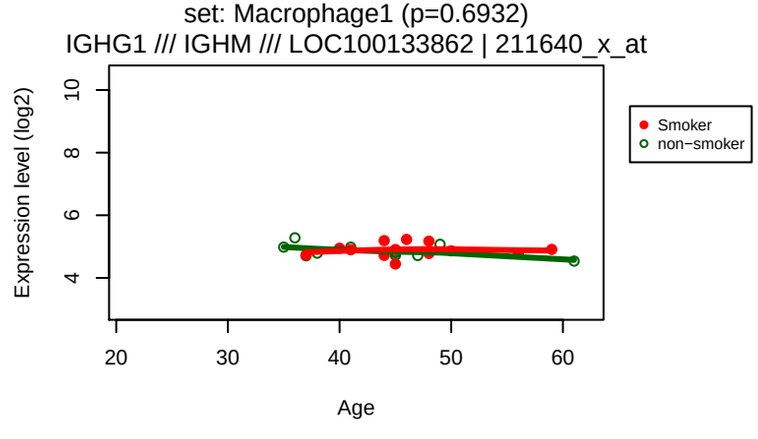

set: SmallAirway1 (p=0.2028)
IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Expression level (log2)
Age
Smoker
non−smoker

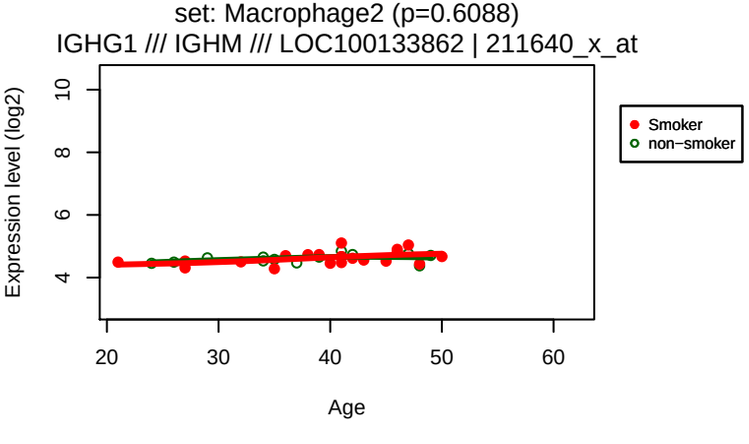

set: SmallAirway2 (p=0.3809)
IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Expression level (log2)
Age
Smoker
non−smoker

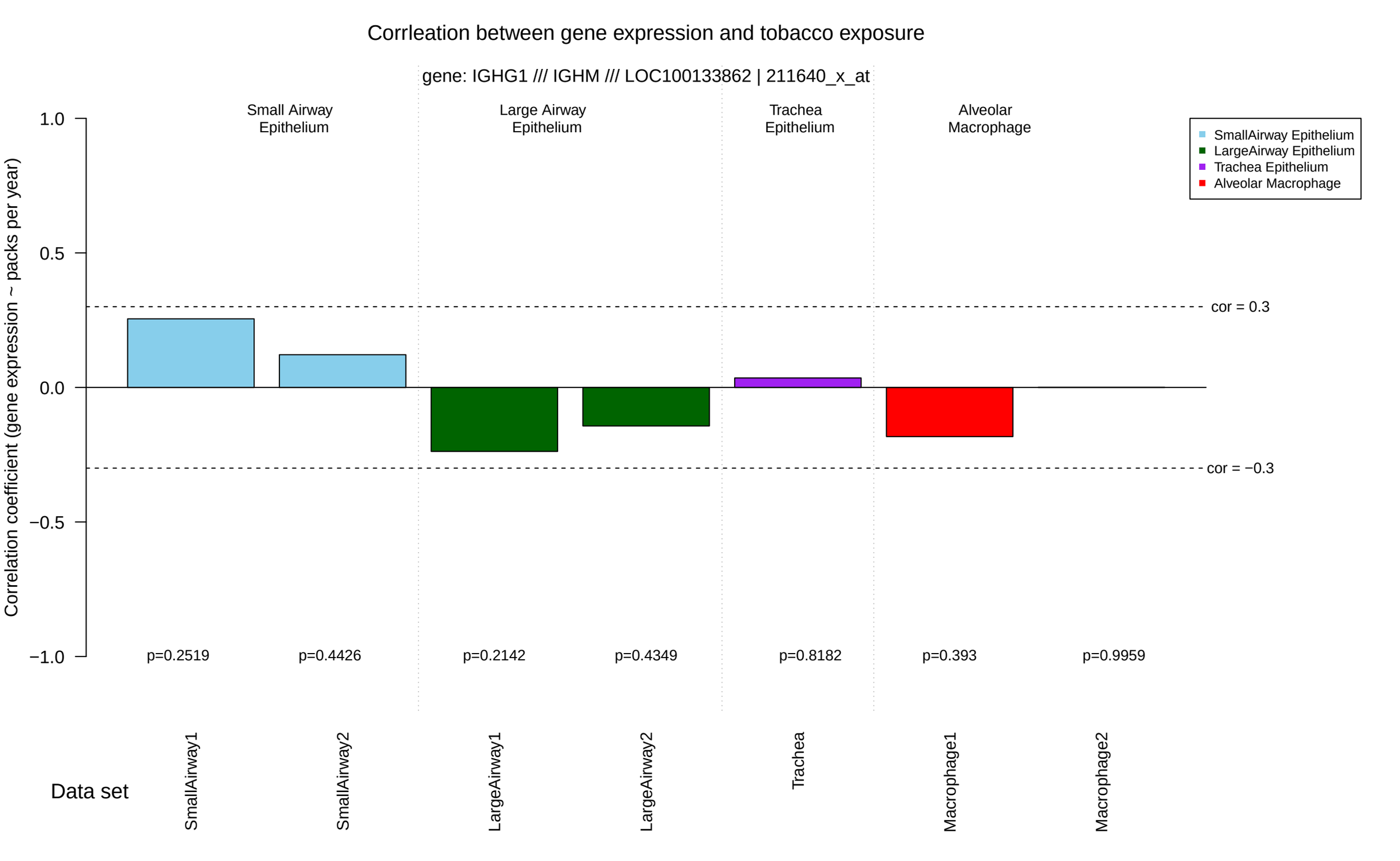

set: LargeAirway1 (p=0.5568)
IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.878)
IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3383)
IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6932)
IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6088)
IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHG1 /// IGHM /// LOC100133862 | 211640_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2519
p=0.4426
p=0.2142
p=0.4349
p=0.8182
p=0.393
p=0.9959
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2